

CSS :placeholder-shown Pseudo-Class Complete Guide to Styling Inputs with Visible Placeholders

CSS Form Styling Mastery

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Chapter 1

Introduction to :placeholder-shown

1.1 What is :placeholder-shown?

The `:placeholder-shown` pseudo-class selects input or textarea elements that are currently displaying placeholder text. It matches when the placeholder is visible (i.e., when the input is empty and has focus or no value).

Key facts about `:placeholder-shown`:

- Targets `<input>` and `<textarea>` elements with placeholder
- Matches only when placeholder is visible (input is empty)
- Requires `placeholder` attribute to work
- Works with input types: text, email, password, number, search, tel, url, etc.
- Does not match when input has value
- Part of CSS Selectors Level 4
- Modern browser support (all modern browsers)
- Enables CSS-only form state detection

1.2 Why Use :placeholder-shown?

`:placeholder-shown` enables elegant form styling without JavaScript:

- **Better UX:** Style inputs based on their state
- **No JavaScript:** Pure CSS form interactions
- **Accessibility:** Improve form usability
- **Visual feedback:** Show input hints and state
- **Clean code:** Simplified form styling logic
- **Performance:** No runtime overhead
- **Responsive states:** Different styling for empty vs filled

1.3 Browser Compatibility

:placeholder-shown is supported in modern browsers:

- Chrome 47+ (2015)
- Firefox 51+ (2017)
- Safari 10.1+ (2017)
- Edge (all versions)
- Opera 34+ (2015)
- Internet Explorer (not supported)

Chapter 2

Basic Syntax and Usage

2.1 Simple :placeholder-shown Usage

```
/* Style input when placeholder is shown */
input:placeholder-shown {
    background-color: #f9f9f9;
    border-color: #ddd;
}

/* Style textarea when placeholder is shown */
textarea:placeholder-shown {
    min-height: 100px;
    background-color: #f5f5f5;
}

/* Combine with other selectors */
input[type="email"]:placeholder-shown {
    border: 1px solid #ccc;
}

input[type="password"]:placeholder-shown {
    letter-spacing: 0;
}
```

2.2 Combining with Other Pseudo-Classes

```
/* When placeholder is shown AND focused */
input:placeholder-shown:focus {
    border-color: #0066cc;
    box-shadow: 0 0 0 3px rgba(0, 102, 204, 0.1);
}

/* When placeholder is shown AND hovered */
input:placeholder-shown:hover {
    background-color: #f0f0f0;
}
```

```
/* Inverse: when NOT showing placeholder */
input:not(:placeholder-shown) {
    background-color: white;
    border-color: #0066cc;
}
```

2.3 HTML Requirements

```
<!-- Required: placeholder attribute -->
<input type="text" placeholder="Enter your name">

<!-- Works with various input types -->
<input type="email" placeholder="Enter email">
<input type="password" placeholder="Enter password">
<input type="search" placeholder="Search...">
<input type="tel" placeholder="Phone number">
<input type="url" placeholder="https://example.com">

<!-- Works with textarea -->
<textarea placeholder="Enter message"></textarea>

<!-- Does NOT work without placeholder -->
<input type="text"> <!-- :placeholder-shown won't match -->
```

Chapter 3

Practical Form Styling

3.1 Floating Label Effect

```
<style>
  .form-group {
    position: relative;
    margin-bottom: 1.5rem;
  }

  .form-group input {
    width: 100%;
    padding: 1rem 0.75rem 0.5rem;
    border: 1px solid #ddd;
    border-radius: 4px;
    font-size: 1rem;
  }

  /* Label positioned absolutely */
  .form-group label {
    position: absolute;
    top: 1rem;
    left: 0.75rem;
    color: #999;
    transition: all 0.3s ease;
    pointer-events: none;
  }

  /* Move label up when input has value */
  input:not(:placeholder-shown) ~ label {
    top: 0.25rem;
    font-size: 0.8rem;
    color: #0066cc;
    font-weight: 600;
  }

  /* Move label up on focus even if empty */
  input:focus ~ label {
    top: 0.25rem;
    font-size: 0.8rem;
```

```
        color: #0066cc;
    }
</style>

<div class="form-group">
  <input type="text" placeholder=" ">
  <label>Your Name</label>
</div>
```

3.2 Conditional Help Text

```
<style>
.form-group {
  margin-bottom: 1.5rem;
}

.form-group input {
  width: 100%;
  padding: 0.75rem;
  border: 1px solid #ddd;
  border-radius: 4px;
}

/* Show help text when placeholder is visible */
.help-text {
  display: none;
  font-size: 0.875rem;
  color: #666;
  margin-top: 0.5rem;
}

input:placeholder-shown ~ .help-text {
  display: block;
}

/* Hide help text when input has value */
input:not(:placeholder-shown) ~ .help-text {
  display: none;
}
</style>

<div class="form-group">
  <input type="email" placeholder="Enter your email">
  <div class="help-text">
    We'll never share your email with anyone else.
  </div>
</div>
```

3.3 Input State Indicators

```
<style>
  .input-wrapper {
    position: relative;
    margin-bottom: 1rem;
  }

  .input-wrapper input {
    width: 100%;
    padding: 0.75rem 2rem 0.75rem 0.75rem;
    border: 2px solid #ddd;
    border-radius: 4px;
    transition: all 0.3s ease;
  }

  /* Icon shown when placeholder is visible */
  .input-icon {
    position: absolute;
    right: 0.75rem;
    top: 50%;
    transform: translateY(-50%);
    font-size: 1.2rem;
    opacity: 0.3;
    transition: opacity 0.3s ease;
  }

  input:placeholder-shown ~ .input-icon {
    opacity: 0.3;
    color: #999;
  }

  /* Icon changes when input has value */
  input:not(:placeholder-shown) ~ .input-icon {
    opacity: 1;
    color: #28a745;
  }

  /* Highlight on focus */
  input:focus {
    border-color: #0066cc;
    box-shadow: 0 0 0 3px rgba(0, 102, 204, 0.1);
  }
</style>

<div class="input-wrapper">
  <input type="email" placeholder="Enter email">
  <span class="input-icon"> </span>
</div>
```

3.4 Placeholder-Aware Labels

```
<style>
```

```
/* When placeholder shown, label is small and inside */
input {
  padding: 0.75rem;
  border: 1px solid #ddd;
  border-radius: 4px;
  width: 100%;
}

label {
  display: block;
  margin-bottom: 0.5rem;
  font-weight: 600;
  color: #333;
}

/* Hide label when placeholder is visible in large inputs */
textarea:placeholder-shown {
  min-height: 100px;
}

textarea {
  width: 100%;
  padding: 0.75rem;
  border: 1px solid #ddd;
  font-family: inherit;
}

/* Different styling based on placeholder state */
input:placeholder-shown {
  background-color: #fafafa;
}

input:not(:placeholder-shown) {
  background-color: white;
  border-color: #0066cc;
}
</style>

<div>
  <label for="name">Full Name</label>
  <input id="name" type="text" placeholder="Enter your full name">
</div>
```

Chapter 4

Advanced Techniques

4.1 Form Validation Styling

```
<style>
/* Base input styling */
input {
    width: 100%;
    padding: 0.75rem;
    border: 2px solid #ddd;
    border-radius: 4px;
    transition: all 0.3s ease;
}

/* Empty state - placeholder visible */
input:placeholder-shown {
    background-color: #f9f9f9;
}

/* Filled state - valid value (no validity pseudo-class needed) */
input:not(:placeholder-shown) {
    background-color: white;
    border-color: #28a745;
}

/* Focus state when empty */
input:placeholder-shown:focus {
    border-color: #0066cc;
    box-shadow: 0 0 0 3px rgba(0, 102, 204, 0.1);
}

/* Focus state when filled */
input:not(:placeholder-shown):focus {
    border-color: #0052a3;
    box-shadow: 0 0 0 3px rgba(0, 102, 204, 0.1);
}
</style>

<input type="email" placeholder="Enter your email">
<input type="password" placeholder="Enter password">
```

4.2 Multi-State Form Groups

```
<style>
  .form-group {
    margin-bottom: 1.5rem;
  }

  .form-group input {
    width: 100%;
    padding: 0.75rem;
    border: 2px solid #ddd;
    border-radius: 4px;
    transition: all 0.3s ease;
  }

  /* Status messages - shown based on input state */
  .status-message {
    font-size: 0.875rem;
    margin-top: 0.5rem;
    display: none;
  }

  /* Show only when placeholder is visible */
  .status-required {
    color: #999;
  }

  input:placeholder-shown ~ .status-required {
    display: block;
  }

  /* Show only when input has value */
  .status-filled {
    color: #28a745;
  }

  input:not(:placeholder-shown) ~ .status-filled {
    display: block;
  }

  /* Styling based on input type */
  input[type="email"]:placeholder-shown {
    background: linear-gradient(to right, #f9f9f9, #fff);
  }

  input[type="email"]:not(:placeholder-shown) {
    background: linear-gradient(to right, #f0f8f0, #fff);
  }
</style>

<div class="form-group">
  <input type="email" placeholder="Enter email address">
```

```
<div class="status-message status-required">Email address  
  required</div>  
<div class="status-message status-filled">Email entered </div>  
</div>
```

4.3 Dynamic Button States

```
<style>  
  form {  
    display: flex;  
    flex-direction: column;  
    gap: 1rem;  
  }  
  
  input {  
    padding: 0.75rem;  
    border: 1px solid #ddd;  
    border-radius: 4px;  
  }  
  
  button {  
    padding: 0.75rem 1.5rem;  
    border: none;  
    border-radius: 4px;  
    font-weight: 600;  
    cursor: pointer;  
    transition: all 0.3s ease;  
  }  
  
  /* Disable submit button when form is empty */  
  button {  
    background-color: #0066cc;  
    color: white;  
  }  
  
  /* When all inputs show placeholder (empty form) */  
  input:placeholder-shown ~ button {  
    background-color: #ccc;  
    cursor: not-allowed;  
    opacity: 0.6;  
  }  
  
  /* When at least one input has value */  
  input:not(:placeholder-shown) ~ button {  
    background-color: #0066cc;  
    opacity: 1;  
    cursor: pointer;  
  }  
  
  button:hover:not(:disabled) {  
    background-color: #0052a3;  
  }  
</style>
```

```
</style>

<form>
  <input type="text" placeholder="Enter name" required>
  <input type="email" placeholder="Enter email" required>
  <button type="submit">Submit</button>
</form>
```

Chapter 5

Real-World Examples

5.1 Example 1: Login Form

```
<style>
.login-container {
  max-width: 400px;
  margin: 2rem auto;
}

.login-form {
  display: flex;
  flex-direction: column;
  gap: 1.5rem;
}

.form-group {
  position: relative;
}

.form-group input {
  width: 100%;
  padding: 1rem 0.75rem 0.5rem;
  border: 2px solid #ddd;
  border-radius: 8px;
  font-size: 1rem;
  transition: all 0.3s ease;
}

.form-group label {
  position: absolute;
  top: 1rem;
  left: 0.75rem;
  color: #999;
  font-size: 1rem;
  font-weight: 500;
  transition: all 0.3s ease;
  pointer-events: none;
}
```

```

/* Move label up when placeholder is shown and focused */
input:focus ~ label,
input:not(:placeholder-shown) ~ label {
  top: 0.25rem;
  font-size: 0.75rem;
  font-weight: 600;
  color: #0066cc;
}

/* Focus styling */
input:focus {
  border-color: #0066cc;
  box-shadow: 0 0 0 3px rgba(0, 102, 204, 0.1);
}

.login-button {
  padding: 1rem;
  background: #0066cc;
  color: white;
  border: none;
  border-radius: 8px;
  font-weight: 600;
  cursor: pointer;
}

/* Button styling based on input state */
input:placeholder-shown ~ .login-button {
  opacity: 0.6;
  cursor: not-allowed;
}
</style>

<div class="login-container">
  <form class="login-form">
    <div class="form-group">
      <input type="email" placeholder=" ">
      <label>Email Address</label>
    </div>
    <div class="form-group">
      <input type="password" placeholder=" ">
      <label>Password</label>
    </div>
    <button type="submit" class="login-button">Sign In</button>
  </form>
</div>

```

5.2 Example 2: Search Form

```

<style>
  .search-wrapper {
    position: relative;
    max-width: 600px;

```

```
        margin: 1rem auto;
    }

    .search-input {
        width: 100%;
        padding: 0.75rem 2.5rem 0.75rem 1rem;
        border: 2px solid #ddd;
        border-radius: 25px;
        font-size: 1rem;
        transition: all 0.3s ease;
    }

    /* Icon styling */
    .search-icon {
        position: absolute;
        right: 1rem;
        top: 50%;
        transform: translateY(-50%);
        font-size: 1.2rem;
        opacity: 0.5;
        transition: opacity 0.3s ease;
    }

    /* Show faded icon when placeholder visible */
    .search-input:placeholder-shown ~ .search-icon {
        opacity: 0.3;
        color: #999;
    }

    /* Show bright icon when searching */
    .search-input:not(:placeholder-shown) ~ .search-icon {
        opacity: 1;
        color: #0066cc;
    }

    /* Enlarge on focus when empty */
    .search-input:focus {
        border-color: #0066cc;
        box-shadow: 0 4px 12px rgba(0, 102, 204, 0.2);
    }

    /* Different styling for active search */
    .search-input:not(:placeholder-shown) {
        border-color: #0066cc;
    }
}
</style>

<div class="search-wrapper">
    <input
        type="search"
        class="search-input"
        placeholder="Search posts, topics, and more...">
    <span class="search-icon"> </span>
</div>
```

5.3 Example 3: Newsletter Signup

```
<style>
  .newsletter {
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
    padding: 2rem;
    border-radius: 12px;
    color: white;
  }

  .newsletter h2 {
    margin-top: 0;
  }

  .newsletter-form {
    display: flex;
    gap: 0.5rem;
  }

  .newsletter-input {
    flex: 1;
    padding: 0.75rem 1rem;
    border: 2px solid rgba(255, 255, 255, 0.3);
    border-radius: 8px;
    background: rgba(255, 255, 255, 0.9);
    font-size: 1rem;
  }

  /* Placeholder visible - subtle styling */
  .newsletter-input:placeholder-shown {
    background: rgba(255, 255, 255, 0.85);
  }

  /* Input has value - highlight */
  .newsletter-input:not(:placeholder-shown) {
    background: white;
    border-color: white;
  }

  .newsletter-button {
    padding: 0.75rem 1.5rem;
    background: white;
    color: #667eea;
    border: none;
    border-radius: 8px;
    font-weight: 600;
    cursor: pointer;
    transition: all 0.3s ease;
  }
```

```
/* Disable when input empty */
.newsletter-input:placeholder-shown ~ .newsletter-button {
  opacity: 0.6;
  cursor: not-allowed;
}

.newsletter-button:hover:not(:disabled) {
  transform: translateY(-2px);
  box-shadow: 0 4px 12px rgba(0, 0, 0, 0.2);
}
</style>

<div class="newsletter">
  <h2>Subscribe to Our Newsletter</h2>
  <form class="newsletter-form">
    <input
      type="email"
      class="newsletter-input"
      placeholder="Enter your email">
    <button type="submit" class="newsletter-button">Subscribe</button>
  </form>
</div>
```

Chapter 6

Accessibility and Best Practices

6.1 Accessibility Considerations

```
/* Ensure focus is visible */
input:focus {
    outline: 2px solid #0066cc;
    outline-offset: 2px;
}

/* Use color + other indicators, not just color */
input:placeholder-shown {
    border-color: #ddd;
}

input:placeholder-shown:focus {
    border-color: #0066cc;
    box-shadow: 0 0 0 3px rgba(0, 102, 204, 0.2);
}

/* Support reduced motion */
@media (prefers-reduced-motion: reduce) {
    input {
        transition: none;
    }
}

/* High contrast mode */
@media (prefers-contrast: more) {
    input:placeholder-shown {
        border: 2px solid #000;
    }

    input:not(:placeholder-shown) {
        border: 2px solid #0066cc;
    }
}

/* Ensure label is associated */
label {
```

```
    display: block;
    margin-bottom: 0.5rem;
    font-weight: 600;
}

input {
    width: 100%;
}
```

6.2 Best Practices

1. **Always include labels:** Use proper <label> elements
2. **Meaningful placeholders:** Use clear, helpful placeholder text
3. **Visible focus:** Always show focus state
4. **Test thoroughly:** Verify in different browsers
5. **Combine with :valid:** Use with validation pseudo-classes
6. **Fallback styling:** Don't depend only on :placeholder-shown
7. **Performance:** :placeholder-shown is efficient, no JavaScript needed

6.3 Common Pitfalls

```
/* WRONG: Relying only on color to indicate state */
input:placeholder-shown { border-color: #ddd; }
input:not(:placeholder-shown) { border-color: #0066cc; }

/* RIGHT: Use multiple indicators */
input:placeholder-shown {
    border-color: #ddd;
    background-color: #f9f9f9;
}

input:not(:placeholder-shown) {
    border-color: #0066cc;
    background-color: white;
    box-shadow: 0 0 0 1px rgba(0, 102, 204, 0.1);
}

/* WRONG: Hiding labels when placeholder shown */
label { display: none; }
input:not(:placeholder-shown) ~ label { display: block; }

/* RIGHT: Always show labels or use aria-label */
label { display: block; }
input { padding: 0.75rem; }
```

Chapter 7

Quick Reference

7.1 Basic Syntax

```
/* When placeholder is visible (input empty) */
input:placeholder-shown { }

/* When placeholder is NOT visible (input has value) */
input:not(:placeholder-shown) { }

/* Combined with focus */
input:placeholder-shown:focus { }
input:not(:placeholder-shown):focus { }

/* Combined with other pseudo-classes */
input:placeholder-shown:hover { }
input:placeholder-shown:disabled { }

/* Combined with attribute selectors */
input[type="email"]:placeholder-shown { }
textarea:placeholder-shown { }
```

7.2 Common Use Cases

```
/* Hide helper text when typing */
.help-text {
    display: block;
}
input:not(:placeholder-shown) ~ .help-text {
    display: none;
}

/* Change input appearance when filled */
input:not(:placeholder-shown) {
    border-color: #28a745;
    box-shadow: 0 0 0 3px rgba(40, 167, 69, 0.1);
}
```

```
/* Show status icon only when filled */
.check-icon {
    display: none;
}
input:not(:placeholder-shown) ~ .check-icon {
    display: inline;
}

/* Disable button when form empty */
button {
    opacity: 0.6;
    cursor: not-allowed;
}
input:not(:placeholder-shown) ~ button {
    opacity: 1;
    cursor: pointer;
}
```

7.3 Complete Form Template

```
/* Production-ready form setup */

.form-group {
    margin-bottom: 1.5rem;
    position: relative;
}

input, textarea {
    width: 100%;
    padding: 0.75rem;
    border: 2px solid #ddd;
    border-radius: 4px;
    font-family: inherit;
    font-size: 1rem;
    transition: all 0.3s ease;
}

/* Focus state */
input:focus, textarea:focus {
    outline: none;
    border-color: #0066cc;
    box-shadow: 0 0 0 3px rgba(0, 102, 204, 0.1);
}

/* Empty state indicator */
input:placeholder-shown {
    background-color: #f9f9f9;
}

/* Filled state indicator */
input:not(:placeholder-shown) {
    background-color: white;
}
```

```
    border-color: #0066cc;
}

/* Help text management */
.help-text {
    font-size: 0.875rem;
    color: #666;
    margin-top: 0.5rem;
    display: none;
}

input:placeholder-shown ~ .help-text {
    display: block;
}

/* Submit button */
button {
    padding: 0.75rem 1.5rem;
    background: #0066cc;
    color: white;
    border: none;
    border-radius: 4px;
    cursor: pointer;
}

input:placeholder-shown ~ button {
    opacity: 0.5;
    cursor: not-allowed;
}

/* Accessibility */
@media (prefers-reduced-motion: reduce) {
    input, textarea {
        transition: none;
    }
}
```